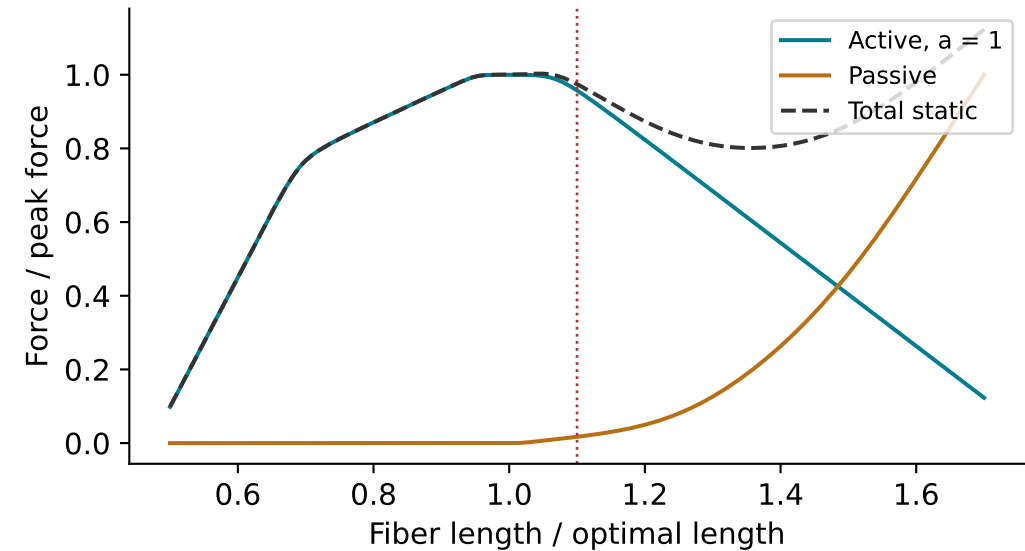


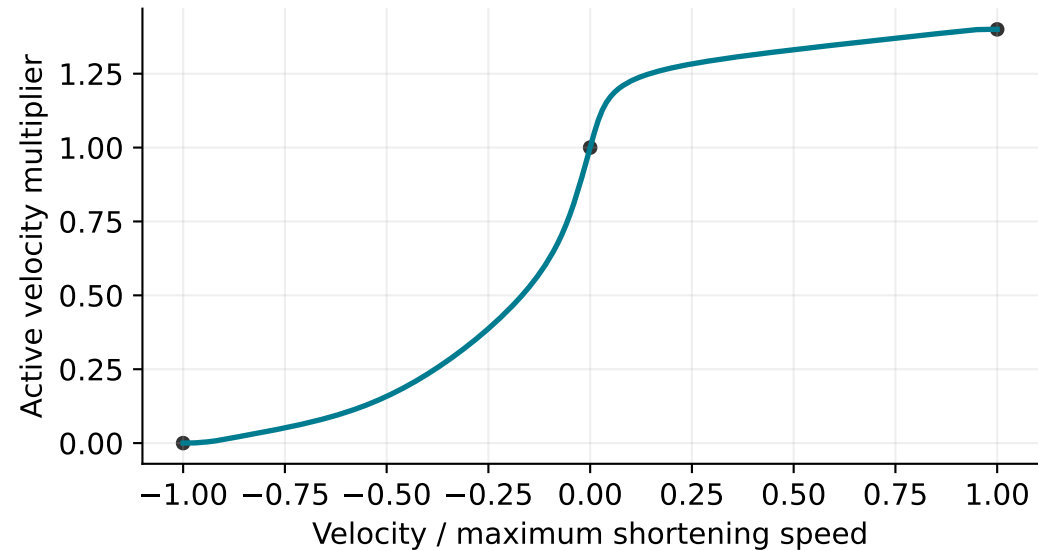
Pinned Millard musculotendon reference cases

OpenSim 4.5.2 kernels · educational scales · zero pennation · compliant tendon

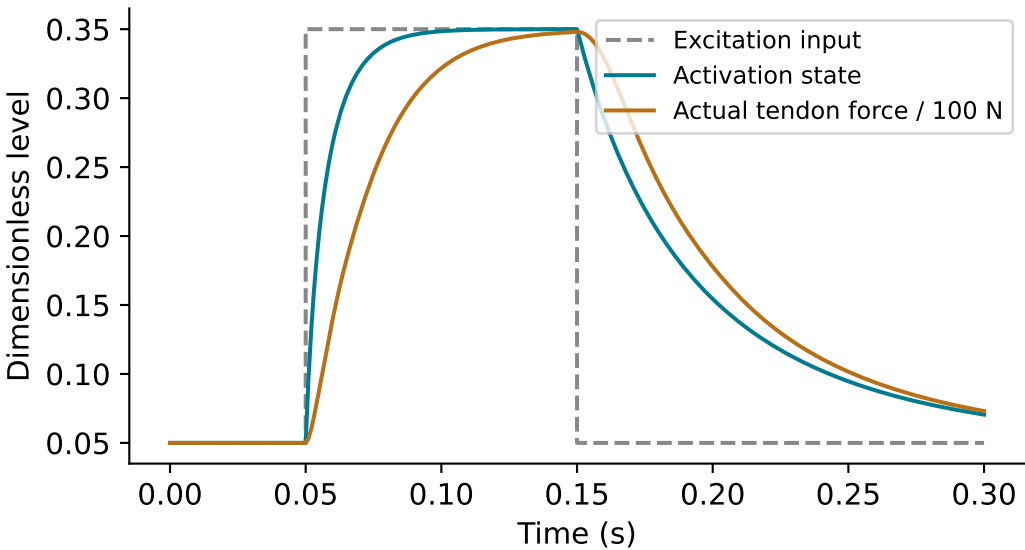
Original force-length curves



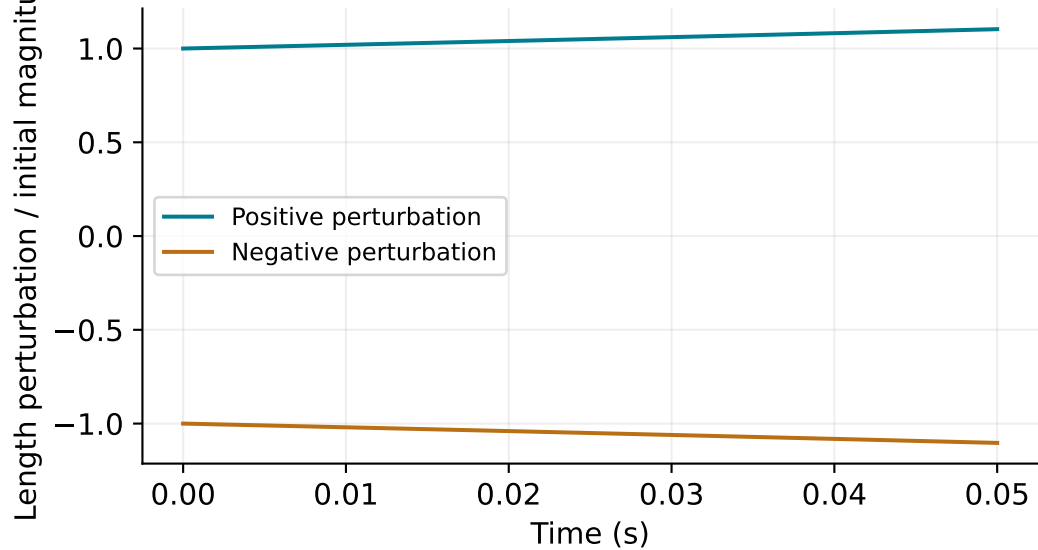
Shortening < 0; lengthening > 0



Held length: force is a response



Fixed force: descending instability retained



Kernel value error 4.14e-15 | matched-time state error 1.39e-10
Growth rate 1.9633 /s | force residual 2.33e-13 N | adaptive work error 2.82e-10 J
Source-kernel and ODE reproduction only; no controller, whole-arm stability, or human calibration claim.